Murat Ambarkutuk

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Professional Experience

R² Robotics - RANPAK **November 2024 – Present**

Principal Perception Engineer *January 2025 – Present*

Sr. Perception Engineer *November 2024 – January 2025*

- Own the architecture and infrastructure of the machine learning and perception stack, development to runtime; the team builds features on its tooling and contracts, cutting development from 6 to 3 sprints for features and 12 to 2 for models.
  - Built r2-bb, the training path: pretrained and froze ViT and ConvNeXt V2 backbones in-house ([**pretrain-images**] images) and defined the contracts that let independent task heads share one backbone pass, amortizing edge inference at industrial scale, cutting p50 latency from [**amort-sep-ms**] ms to [**amort-shared-ms**] ms for [**amort-k**] heads on Orin Nano, with [**amort-mem-pct**]% less memory and [**amort-energy-pct**]% less energy per frame.
  - Built libinference, the inference path: a header-only ONNX/TensorRT library for Linux amd64/arm64 and Jetson (Nano, Orin Nano, NX) with dynamic shapes, multi-GPU support, and CUDA stream parallelism, running on [**fleet-devices**] devices across [**fleet-sites**] sites, with [**libinf-overhead**]% overhead over raw TensorRT and [**libinf-load-s**] s engine load on Orin Nano.
  - Optimized models with automatic mixed precision (AMP), pruning-aware training (PAT), post-training quantization (PTQ), and knowledge distillation from in-house or foundation-encoder teachers ([**comp-size-x**]× smaller, [**comp-speed-x**]× faster, [**comp-acc-pt**] pt accuracy cost).
  - Built synthetic data generation pipelines with NVIDIA Isaac Sim and a Blender-based native generator, [**synth-images**]k images lifting held-out real accuracy [**synth-lift**] pt.
- Architect Decision Tower 2.0, the end-of-line packaging perception stack rewritten on r2-bb with amortized inference, sustaining [**dt-ipm**] items per minute per station.
  - Designed every model as a head on the shared backbone instead of a network of its own: classifier, heatmap keypoint, Mask R-CNN, and oriented-box heads ([**heads-prod**] in production, [**heads-dev**] in development); [**cls-f1**] classifier F1, [**kp-oks**] keypoint OKS, [**seg-miou**] segmentation mIoU.
  - Proved it on teammates' models: two ported their HRNet and YOLO into its keypoint and Mask R-CNN heads.
  - Fine-tuned and deployed its SigLIP classifier with independent sigmoid heads; built multi-camera calibration, 3D reconstruction, and neural scene representations ([**dt-reproj**] px reprojection error, [**dt-recon-mm**] mm reconstruction error).
- Build the MLOps stack for the ML lifecycle (AMP training, PAT, PTQ, deployment): MLflow (PyTorch, TensorFlow, Keras), a private S3-backed APT repository with release and dev channels, and libinference's release ceremonies, shipping [**rel-per-month**] releases a month across [**apt-channels**] channels.
- Own the project board for a six-person team that operates ten-strong through agentic development and review pipelines ([**velocity-x**]× more merged changes per week): divide products into MVPs ([**mvps-shipped**] shipped), develop technical plans, run agile with weekly demos, and mentor the team.

Baykar Defense (TR) - Atlas Imaging (US) **2018 – 2020**

Computer Vision Engineer *Remote*

- Developed orthorectified maps from autonomous-UAV aerial photos: derived the pose of the UAV and its components from distributed INS measurements, then iteratively projected image corners onto terrain with elevation changes.
- Built a statistical sensor fusion stage that corrects the results of the earlier map-making stages.
- Introduced test-driven development and architected the CI/CD pipeline that runs and reports the tests automatically.

Education

PhD in Computer Engineering **2018 – 2024**

The Bradley Department of Electrical and Computer Engineering at Virginia Tech *Blacksburg, VA*

- Paul E. Torgersen Research Excellence Award
- The Outstanding Research Award and Fellowship from the Turkish Ministry of National Education

Dissertation: A Sensor Fusion Technique for Gait Analysis from Structural Vibration and Computer Vision

MS in Mechanical Engineering **2015 – 2018**

Mechanical Engineering at Virginia Tech *Blacksburg, VA*

Thesis: A Grid based Indoor Radiolocation Technique Based on Spatially Coherent Path Loss Model

BS in Mechatronics Engineering **2008 – 2013**

Mechatronics Engineering Department at Kocaeli University *Kocaeli, Turkey*

Thesis: Box Dimensioning System with Kinect Sensor

Technical & Language Skills

Languages: Python, C++, C, CUDA, MATLAB
ML/CV: PyTorch, TensorFlow, Keras, OpenCV, scikit-learn, scikit-image, YOLO, DINOv3, ViT, SigLIP, ConvNeXt V2
3D Vision & Robotics: COLMAP, Multi-view Geometry, TSDF Fusion, 3D Gaussian Splatting, ROS
Simulation & NVIDIA Stack: Isaac Sim (Omniverse), Jetson, Blender, Synthetic Data Generation, CUDA, TensorRT
Optimization & MLOps: ONNX, MLflow, CUDA Kernels, Quantization (PTQ), Pruning (PAT), AMP, Distillation
Tools & Technologies: Linux (amd64, arm64), APT, AWS S3, Git, GitHub, GitLab, CI/CD, Docker, LaTeX
Language Proficiency: Turkish (Native), English (Proficient), German (Novice)

Projects

İşaret: Agent-Run AI & Robotics Newsletter

oruntu | isaret.ai **May 2026 – Present**

- Built and operate an AI-agent pipeline that curates arXiv papers and industry news, writes Turkish summaries with source links and license-credited figures, runs automated review, and publishes each issue; 50+ issues since May 2026.
- Deliver every issue by email, Atom/RSS, a JSON API, LinkedIn, and Telegram; Cloudflare-hosted, GitHub-published.

Sensor Fusion for Occupant Localization in Smart Buildings

Advisors: P Plassmann, CF Jones | Related publications: [1–5] **2018–2024**

- Developed a robust algorithm that estimates heel-strike locations from the accelerometer measurements of a vibrating floor.
- Built a computer vision algorithm that localizes heel-strike locations from stereoscopic camera images.
- Designed a sensor fusion framework that combines the vibration- and vision-based heel-strike locations.
- Created a combined simulation environment that generates the vibration and visual data needed to validate the algorithms.
- Quantified the efficacy and accuracy of the implementations with simulations and controlled experiments.

A Grid-based Indoor Radiolocation Technique based on Spatially Coherent Path Loss Model

Advisor: T Furukawa | Publications: [6] **2016–2017**

- Derived a probabilistic framework for passive radio localization under multipath propagation that determines the location of occupants, robots, and inventory in indoor environments.
- Developed the localization algorithm based on the probabilistic framework.
- Validated the algorithm with controlled experiments whose results were presented at an IEEE RAS conference.

Relevant Publications

[1] **M. Ambarkutuk** and P. E. Plassmann. Modeling and analysis of dispersive propagation of structural waves for vibro-localization. *Sensors*, 24(23):7744, 2024. URL https://doi.org/10.3390/s24237744.

[2] **M. Ambarkutuk**, S. Alajlouni, P. A. Tarazaga, and P. E. Plassmann. A multi-sensor stochastic energy-based vibro-localization technique with byzantine sensor elimination. *Sensors*, 23(23):9309, 2023. URL https://doi.org/10.3390/s23239309.

[3] **M. Ambarkutuk** and P. Tarazaga. Uncertainty analysis of an occupant localization technique based on simulated structural vibrations. In *Dynamics of Civil Structures, Volume 2: Proceedings of the 39th IMAC*. Springer, 2022.

[4] S. Alajlouni, V. V. S. Malladi, and **M. Ambarkutuk**. Investigation of using log-spectrum averaging (cepstral averaging) for blind reconstruction of an unknown impact input force. In *Dynamics of Civil Structures, Volume 2: Proceedings of the 39th IMAC*. Springer, 2022.

[5] S. Alajlouni, **M. Ambarkutuk**, and P. Tarazaga. Towards the detection and localization of multiple occupant footsteps from vibroacoustic measurements. In *Dynamics of Civil Structures, Volume 2: Proceedings of the 38th IMAC, A Conference and Exposition on Structural Dynamics 2020*, page 193. Springer, 2020.

[6] **M. Ambarkutuk** and T. Furukawa. A grid-based indoor radiolocation technique based on spatially coherent path loss model. In *2017 IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI)*, pages 220–226. IEEE, 2017.

[7] H. Ocak, **M. Ambarkütük**, G. Küçükyıldız, and S. Karakaya. Image processing based package volume detection with kinect. In *2015 23nd Signal Processing and Communications Applications Conference (SIU)*, pages 515–518. IEEE, 2015.

Community Service & Leadership

Graduate Student Ambassador, ECE at Virginia Tech · Student Member & Reviewer, IEEE Robotics and Automation Society · Student Member, Society of Experimental Mechanics · Secretary, Turkish Student Association at Virginia Tech